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Fire risk mitigation underpins durable Nature-based Climate Solutions in the Amazon

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Fire-conditioned degradation now rivals deforestation as an independent source of carbon loss across Amazonia. We argue that fire and degradation are first-order risks to Nature-based solutions and must be embedded explicitly in monitoring, reporting and verification (MRV), as well as crediting and jurisdictional finance.

Amazonia's fire regime has undergone a fundamental shift over the past two decades, from episodic, edge-bound ignitions tied to clearing, to drought-conditioned landscape fires that now penetrate intact forest cores. Landscape fires are increasingly decoupled from contemporaneous deforestation[1]. This is a change in system dynamics – stronger positive feedback, lowered recovery potential, and new thresholds – rather than a simple rise in counts or burned area.

More frequent and intense droughts slow vegetation recovery and increase dead biomass[2]; degradation and fragmentation reduce evapotranspiration and moisture recycling[3]; and gaps in policy, enforcement, and provision of fire-free agricultural alternatives – despite advances such as Brazil's National Policy on Integrated Fire Management and growing cross-border cooperation – undermine compliance and accountability[4]. Under current hot, dry conditions, even locally managed or agricultural fires can escape into community-managed forests and extractive reserves, as illustrated by the extensive 2024 burning reported in the Tapajós–Arapiuns Extractive Reserve in Pará, a protected area used by traditional and riverine communities. Collectively such events ratchet the rainforest ecosystem toward persistent, fire-prone states[5].

We argue that Amazonian Nature-based Climate Solutions require a fire-conditioned architecture: one that treats ignition, degradation and post-fire recovery failure as central permanence risks rather than residual uncertainties. Policy should therefore prioritize prevention finance, degradation-explicit reporting, strengthened land tenure and Indigenous stewardship alongside conventional deforestation control.

Reduced resilience, slow recovery

Observed trends show that declining clear-cut deforestation does not necessarily imply declining forest carbon loss: fire-related degradation is becoming a rival source of carbon emissions to the atmosphere. According to Brazil's official monitoring reports, gross deforestation in the legal Amazon fell by roughly half between 2022 (11,600 km^2) and 2025 (5800 km^2, preliminary), with smaller declines in neighbouring countries. Yet fire-conditioned degradation has advanced: alerts from the Brazilian National Institute for Space Research's (INPE) near-real-time deforestation and degradation monitoring system, DETER (Detecção do Desmatamento em Tempo Real), together with complementary analyses, indicate that in recent years forest-fire degradation has affected an area on the order of ten times annual clear-cut deforestation[6]. Emissions from fire-conditioned degradation now approach those from deforestation (roughly 0.05–0.20 Pg C $year^{-1}$ versus 0.06–0.21 Pg C $year^{-1}$), because of the cumulative effects of understory fires: they directly reduce live carbon stocks as well as open canopies; suppress dry-season evapotranspiration; and lock landscapes into low-biomass attractors that recover slowly, if at all[7].

The slow recovery threatens the resilience of the Amazon Basin to act as a carbon sink and potential source of Nature-based Climate Solutions[8,9]. The scientific rationale for Nature-based Climate Solutions rests on large standing carbon stocks, the capacity of intact and recovering forests to store and sequester carbon, and the permanence of those sinks. Fire directly undermines permanence and additionality: if fires and associated degradation erode resilience faster than avoided deforestation and restoration can rebuild it, Nature-based Climate Solutions become ineffective or even counterproductive. Amazon rainforests typically lack the large, dedicated post-fire restoration investments that are common in temperate regions; recovery is often multi-decadal to centennial when it occurs. Thus, temporary climate anomalies or brief governance lapses can produce long-lasting structural losses.

Destabilised water balance

Multiple lines of evidence point to growing systemic vulnerability of rainforests[8]. Extensive deforestation and degradation over the past 40 years have significantly increased surface air temperature, impacted evapotranspiration and reduced precipitation by more than 15 mm during the dry season[10]. When upwind evapotranspiration falters, moisture recycling can be propagated downwind by the reduction of atmospheric moisture transport, resulting in a positive feedback mechanism and self-amplified forest transition, accounting for approximately one-third of all tipping point events[11]. Early-warning diagnostics reveal critical slowing down—rising temporal autocorrelation and variance—across drought-sensitive sub-basins, with drought intensity emerging as the dominant predictor of recovery failure in 2001–2019 records[8,12].

These changes in hydrological balance combined with recurrent and extreme droughts can outpace adaptive capacity of the Amazon forests. Process-based projections that resolve competition among trees, grasses and fire reproduce abrupt forest-to-savanna transitions under high-emissions scenarios; grass-fire feedback then inhibits recovery even after rainfall returns[9]. In short, fire amplifies short-term shocks into persistent regime shifts, nullifying Nature-based Climate Solutions additionality and durability, unless explicitly addressed.

Human ignition as policy leverage

Contrary to framing fires as an inevitable climatic outcome, ignition in the Amazon is overwhelmingly anthropogenic: land clearing, pasture management, and accidental escapes predominate[9]. Yet, deliberate in-forest ignitions linked to illegal clearing and land appropriation also contribute, making deterrence, rapid suppression, and enforceable accountability central to permanence.

Contemporary fire seasons (Fig. 1) are increasingly decoupled from new clear-cutting; for example, only about 19% of January–June 2023 fires coincided with recent deforestation alerts, with most ignitions starting in pastures and degraded forests and then spilling into intact stands under hot, dry conditions[13]. During severe droughts, a non-negligible fraction of ignitions is set directly within forests – including public and protected lands – consistent with illegal land appropriation and fire-enabled deforestation. Fragmentation desiccates edges, raises vapour-pressure deficit and canopy dryness, increasing ignition likelihood and shortening recovery windows. Atmospheric transport amplifies local disturbances basin-wide – empirical analyses indicate average doubling of impact across the basin with hotspots seeing amplifications up to about 250%. As a result, local fires can have far-reaching consequences[14].

This evidence identifies where policy leverage is highest: edges, previously burned forests, degraded belts and chronic ignition sources. Because human actions drive most ignitions and because edge amplification and atmospheric spill overs magnify effects, targeted governance—prevention, ignition control, rapid suppression, and landscape management—can substantially reduce risk.

Fire-risk diagnostics for verification

Nature-based Climate Solutions integrity requires treating fire and degradation as first-order risks in Monitoring Reporting and Verification (MRV)

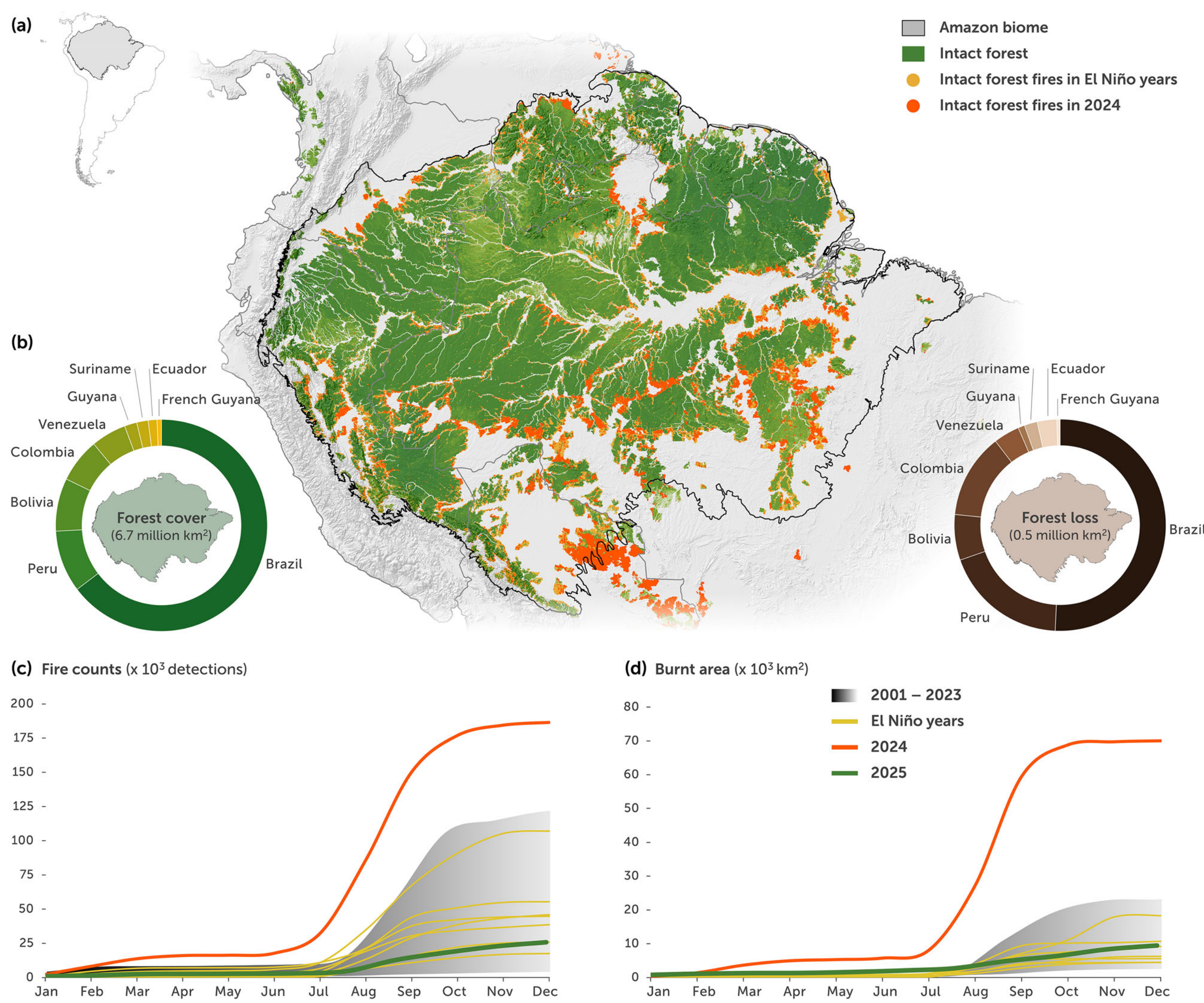

Fig. 1 | Forest fire anomalies in 2024–2025 across the Amazon biome. **a** Amazon biome and high-integrity/conserved forest mask (green mask) with locations of forest fires intersecting this mask in El Niño years (yellow) and in 2024 (orange). **b** Country shares of remaining forest cover and cumulative forest loss since 2001. **c**, **d** Cumulative active-fire detections ($\times 10^3$) and burned area ($\times 10^3\,km^2$) by year: individual 2001–2023 trajectories (grey), El Niño years (yellow), 2024 (orange) and 2025 (green).

systems for credit issuance, finance and governance. Three operational diagnostics can be embedded in jurisdictional policies and markets:

- Anomaly-normalized fire metrics. Report active-fire and burned-area departures relative to a baseline (e.g. 2001–2023) climatology as percentile ranks and distance to a historical minimum envelope (e.g., p5). This isolates anomalous burning beyond expected weather and supports weather-attribution components in jurisdictional performance metrics.
- A decoupling index. Measure the proportion of fires and burned area outside recent clear-cuts to distinguish degradation-driven impacts from conversion. This prevents crediting leakage where avoided deforestation is maintained but degradation increases.
- An enforcement-elasticity coefficient. Quantify how deforestation, burned area and emissions respond to enforcement actions (infractions, fines, seizures). Linking credits to governance performance creates accountability and aligns financial incentives with on-the-ground deterrence.

Embedding these metrics must include social safeguards and equitable implementation: we suggest to stratify indicators by tenure and livelihood class (indigenous lands, settlements, smallholders, protected areas); to pair restrictions with financed fire-free alternatives; providing training to ensure agricultural productivity and food security; and to allocate compensation using jurisdictional indicators. Within jurisdictional REDD+ frameworks—reducing emissions from deforestation and forest degradation, plus conservation, sustainable forest management and enhancement of forest carbon stocks—and regional funds, such metrics enable ex-ante public finance for patrol capacity and ex-post disbursements proportional to verified reductions in illegal ignition and edge desiccation[15].

Prevention as permanence policy

A durable, fire-conditioned Nature-based Climate Solutions architecture integrates climate physics, landscape ecology and law. Policy priorities to secure Nature-based Climate Solutions durability may include:

- Prevention-first ignition control. Institute predictable, graduated zero-ignition rules during high-risk windows triggered by sub-seasonal forecasts, combined with pre-announced enforcement surges and rapid on-site adjudication. Participatory fire-use governance can avoid criminalizing subsistence burning while tightening controls on high-risk corridors[16].
- Degradation-explicit Monitoring Reporting and Verification (MRV) and reporting. Near-real-time detection of understory fires, selective logging, and edge effects should feed jurisdictional performance dashboards. Payments and market access should reward verified reductions in anomaly-normalized fire metrics, not just reduced clearing.
- Market and supply-chain rules. Move from "deforestation-free" to "degradation-free" sourcing for beef, soy, timber and high-risk commodities—geolocated, audited, and jurisdictionally conditioned—so markets penalize fire-enabled degradation.
- Remove frontier-opening incentives. Withdraw or condition projects and policies (road upgrades, permissive land-regularization, new extractive licenses) that increase access rents and ignition pressure along supply chains.
- Scale prevention finance. Prioritize ex-ante funding for brigades, territorial monitoring, ignition buffers, fuel breaks, community training and fire-free land-preparation alternatives. Prevention is cost-effective when unpriced reversal costs (carbon, health, infrastructure, biodiversity) are considered.
- Strengthen tenure and indigenous stewardship. Secure and accelerate titling, fund Indigenous brigades and monitoring, and pair territorial rights with supported prevention capacity – investments that function both as protection and as restoration policy.

Treating fire as a central risk preserves carbon, biodiversity and the Amazon's hydro-ecological functions. Fire-driven degradation shifts communities toward pioneer and fire-tolerant species and reduces richness, often approaching the biodiversity loss seen with conversion in human-modified landscapes[17]. Recurrent fires and edge creation also delay wet-season onset and lengthen dry seasons; because the Amazon recycles roughly half its rainfall, these changes can reduce downwind precipitation and threaten urban water security.

A sceptical test of the 2024–2025 wildfire contrast (Fig. 1) suggests 2025's unusually low season is not explained by weather alone. Basin-mean June-to-September moisture anomalies were near-climatology in 2025 but substantially drier and hotter in 2024 (consistent with a late-phase El Niño and sustained in part by anomalously warm Tropical North Atlantic sea-surface temperatures that can suppress regional moisture availability). Although the weather helped keep fires at bay in 2025, the larger signal is policy preparedness: the severe 2024 season triggered state plans, expanded brigades, targeted burn bans and tighter supply-chain scrutiny.

The 2025 outcome – with fire activity well below the 2001–2023 interquartile range – indicates that decisive governance, combined with modestly supportive weather, can suppress ignition at scale. This example supports our conclusion that fire risk is tractable and that prevention-first policies can make nature-based solutions durable.

Amazonian Nature-based Climate Solutions will remain fragile unless fire and degradation are treated as first-order risks. The contrast between catastrophic and suppressed seasons in the past few years shows both the threat and the remedy: fires can rapidly erase mitigation gains, yet targeted governance can stabilize the basin's carbon, water and biodiversity.

A science aligned Nature-based Climate Solutions architecture must prioritize drought window ignition control; embed degradation-explicit Monitoring Reporting and Verification; finance and fund Indigenous stewardship; and remove frontier-opening incentives. Only by integrating prevention, monitoring, market rules and tenure security can Nature-based Climate Solutions shift from fragile accounting to durable resilience.

Data availability

All the data used in this study are publicly available.

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Author contributions

M.H. conceptualized the manuscript, led the literature synthesis and interpretation, prepared the original draft and figure, coordinated revisions, and prepared the final submitted version. S.S.S. conceptualized the manuscript, secured funding, contributed to the original draft, and provided scientific guidance and critical revisions. A.A.C.A., L.E.O.C.A., L.O.A., C.H.L.S.-J., P.A. and C.A.N. contributed to scientific discussion and interpretation, revised the manuscript, and approved the final version.

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